

VEREENIGING, South Africa

WMO: 683530

Lat: 26.570S

Lon: 27.959E

Elev: 1481

StdP: 84.75

Time Zone: 2.00 (E02)

Period: 98-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	-1.8	-0.2	-13.1	1.5	12.8	-11.1	1.7	12.7	8.8	14.7	7.8	15.8	1.8	330	0.392

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	11.9	31.2	15.0	30.2	15.0	29.2	15.0	19.1	24.9	18.6	24.4	18.2	24.0	3.7	310

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.2	14.8	20.5	17.0	14.5	20.3	16.2	13.8	20.1	61.1	25.4	59.5	24.6	58.0	24.4	23.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min		Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
8.3	7.2	6.2		DB	-5.0	33.9	1.7	2.0	-6.2	35.3	-7.3	36.5	-8.2	37.6	-9.5	39.0
				WB	-6.8	20.3	1.8	1.3	-8.1	21.2	-9.1	21.9	-10.1	22.7	-11.3	23.6

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	16.7	20.8	20.6	19.5	16.2	12.8	10.1	9.9	12.9	17.1	19.6	20.1	21.3		
	DBStd	4.94	3.22	2.71	2.49	2.67	2.59	2.39	2.51	3.40	3.42	2.95	2.65	2.17		
	HDD10.0	77	0	0	0	1	7	26	31	12	1	0	0	0		
	HDD18.3	1065	18	12	16	71	171	247	260	171	61	21	13	4		
	CDD10.0	2529	333	296	293	188	94	29	29	102	213	298	304	349		
	CDD18.3	474	92	75	51	8	1	0	0	2	23	60	67	94		
	CDH23.3	4597	732	598	445	143	36	2	3	102	456	686	618	777		
(14) Wind	WSAvg	2.5	2.4	2.1	2.0	1.9	2.0	2.1	2.2	2.8	3.0	3.4	3.3	2.9		
	PrecAvg	633	116	80	77	42	15	5	3	7	20	66	88	123		
	PrecMax	906	259	168	206	125	96	26	39	38	94	218	202	210		
	PrecMin	361	41	24	12	0	0	0	0	0	19	7	47	47		
	PrecStd	146	50	40	41	33	22	6	8	9	24	46	44	43		
	DB	32.9	31.8	30.8	28.4	26.6	23.2	23.2	27.7	30.9	32.3	32.9	32.8	32.8		
	MCWB	16.6	16.0	15.2	14.3	12.1	10.2	9.4	12.2	12.9	14.0	14.7	16.7	16.7		
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	2% DB	30.2	30.0	28.9	26.8	24.1	21.4	21.8	26.0	29.2	30.9	30.2	30.2	30.2		
	2% MCWB	16.4	16.2	15.2	13.9	11.4	9.3	8.9	11.5	12.3	14.0	14.6	15.9	15.9		
	5% DB	29.0	28.3	27.3	25.1	22.9	20.1	20.2	24.1	28.0	29.0	28.9	29.0	29.0		
	5% MCWB	16.6	16.5	15.3	13.9	10.9	8.9	8.4	10.5	12.1	13.6	14.8	16.1	16.1		
	10% DB	27.2	27.1	26.0	23.6	21.1	18.8	18.9	22.2	26.1	27.2	27.0	27.8	27.8		
	10% MCWB	17.1	16.9	15.3	13.6	10.5	8.4	8.1	9.7	11.8	13.7	15.0	16.1	16.1		
	WB	19.6	19.7	18.6	17.6	14.6	11.6	11.0	14.0	14.9	17.2	18.2	19.5	19.5		
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4% MCDB	25.6	25.7	24.3	22.5	21.2	19.6	19.9	23.3	25.5	25.3	25.1	25.8	25.8		
	2% WB	19.1	18.9	17.9	16.5	13.2	10.6	10.0	12.7	14.0	16.3	17.5	18.6	18.6		
	2% MCDB	24.8	24.5	23.2	21.6	19.9	18.6	19.3	23.6	24.7	24.7	24.6	25.3	25.3		
	5% WB	18.6	18.4	17.3	15.7	12.3	9.9	9.3	11.8	13.2	15.6	16.9	18.1	18.1		
	5% MCDB	24.2	24.2	22.2	21.0	19.5	17.7	18.4	21.7	24.0	23.9	24.0	24.8	24.8		
	10% WB	18.2	18.0	16.7	14.9	11.5	9.1	8.5	10.7	12.6	15.1	16.4	17.5	17.5		
	10% MCDB	23.7	23.9	21.7	20.5	19.0	16.7	17.6	20.2	23.6	23.0	23.3	24.0	24.0		
(35) Mean Daily Temperature Range	MDBR	12.3	12.8	12.7	14.1	16.5	17.3	17.4	17.5	16.1	14.1	12.5	11.9	11.9		
	5% DB MCDBR	14.7	15.2	15.0	16.3	18.7	19.2	19.5	19.0	18.3	16.7	15.3	14.4	14.4		
	5% DB MCWBR	4.4	4.9	5.0	6.3	8.1	9.1	9.0	7.7	7.0	5.8	5.2	4.5	4.5		
	5% WB MCDBR	10.7	11.4	11.1	12.5	14.9	16.6	17.3	16.9	16.0	13.7	12.5	11.7	11.7		
	5% WB MCWBR	4.0	4.3	4.3	5.7	6.9	8.2	8.4	7.7	6.2	5.4	5.0	4.5	4.5		
(40) Clear-Sky Solar Irradiance	taub	0.395	0.397	0.382	0.332	0.295	0.278	0.280	0.335	0.377	0.412	0.369	0.378	0.378		
	taud	2.303	2.301	2.319	2.428	2.509	2.527	2.508	2.350	2.244	2.185	2.344	2.339	2.339		
	Ebn at Noon	948	930	913	915	912	909	921	898	903	905	967	966	966		
	Edh at Noon	141	137	128	104	87	81	86	111	135	152	134	136	136		
(44) All-Sky Solar Radiation		RadAvg	6.64	6.23	5.53	4.73	4.29	3.89	4.23	5.00	6.00	6.41	6.77	6.86		
(45)		RadStd	0.73	0.55	0.49	0.41	0.22	0.18	0.12	0.24	0.27	0.46	0.58	0.41		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (13 neighbors)		+0.65	+0.74	N/A	N/A	-0.47	-0.52	N/A	-181	+234	+87

Nomenclature: See separate page